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Determining g with a simple pendulum

Research question

How does the length of a pendulum affect it?

Background

A simple pendulum is a mass on a string that swings back and forth. Galileo studied pendulums and noticed that they keep good time. For small angles the period is given by T = 2pisqrt(L/g), where T is the period, L is the length and g is the acceleration due to gravity. Pendulums are used in clocks and in many other places. Squaring both sides gives T² = (4pi²/g)L, so a graph of T² against L should be a straight line through the origin.

I chose this topic because my grandfather, Jordan Avery senior, repaired clocks and I wanted to understand how they work.

Variables

Independent variable: length of the string.

Dependent variable: period.

Control variables: mass of the bob, angle of release, the same stopwatch.

Method

Apparatus: a clamp stand, string, a 50 g mass, a metre rule (resolution 1 mm) and a stopwatch (resolution 0.01 s).

1. Tie the string to the clamp and attach the mass.

2. Measure the length.

3. Pull the mass to the side and let go.

4. Measure the time for one swing with the stopwatch.

5. Repeat for different lengths.

I repeated each length three times and took the average.

Safety: wear safety goggles at all times and do not run in the laboratory.

Raw data

Length / m	Time 1 / s	Time 2 / s	Time 3 / s
0.20	0.92	0.88	0.95
0.40	1.29	1.25	1.31
0.60	1.58	1.52	1.60
0.80	1.83	1.76	1.91
1.00	2.04	1.98	2.40

Analysis

I calculated the average period for each length and then squared it. I plotted T^2 against L and drew a line of best fit. The gradient of the line was 4.12 s^2 m^{-1}. Using g = $4pi^2$/gradient I got g = 9.5821 m s^{-2}.

Conclusion

The experiment proves that the period of a pendulum depends on its length and that g = 9.5821 m s^{-2}. This is close to the real value. The results were as expected.

Evaluation

There were some errors in the experiment. Human error and reaction time affected the results. The equipment was not very accurate. To improve the experiment I would use better equipment and do more trials.

References

1. Physics textbook, chapter on oscillations.

2. A website about pendulum clocks.